

Chapter 14 — Sutures and Suturing

- **Suture material, size, needle type and technique** are chosen for the specific anastomosis, repair or closure required.
- Three primary determinants: **tissue type** (skin, fascia, soft tissue, mucosa, viscera), **anticipated tension on the suture**, and **required duration of tensile strength**.
- Most materials are decades old; **barbed sutures** are the main recent innovation. Knowing each suture's and needle's strengths and weaknesses reduces complications.

NEEDLE TYPES

- **The application dictates the needle**; most sutures come with a range of needles, so the **needle is chosen independently of the suture**. Needles vary in **point, diameter, arc and radius** to suit tissue and operative space.

Needle Point

- Broad division: **non-cutting versus cutting** — the examinable pair:
 - **Non-cutting** (taper and blunt): tapers from a wider body to a narrow tip; **stretches the tissue**, which then retracts around the suture — less trauma.
 - **Cutting**: **cuts as it passes** — passes more easily, at the cost of slightly more trauma.

NEEDLE POINTS COMPARED (FIG. 14.1A–D)

Point	Shaft and tip	Behaviour in tissue	Typical use	Named needles
Taper	Round shaft, tapers to sharp tip	Stretches tissue before the suture enters; atraumatic with modest effort — most commonly used	General and delicate tissue	RB/CV, SH, CT (½ circle); UR/GU (5/8 circle)
Blunt	Round shaft, blunt tip	Harder to pass, needs more force; minimises needlestick injury to operating room staff and patient	Non-dense tissue, e.g. subcutaneous fat	BP (ballpoint), CTB (circle taper blunt) — both ½ circle; BP larger diameter
Cutting	Triangular; cutting edge on concave inner curve	Cuts in the direction of pull; more traumatic but passes dense and scarred tissue easily	Skin closure	V series: V-26, V-4 (3/8 circle); V-5, V-7, V-34, V-37, V-40, V-56 (½ circle)
Reverse cutting	Triangular; cutting edge on convex outer curve	Avoids accidental shearing of tissue on pull-through, but leaves larger holes than a conventional cutting needle	Retention sutures in complex abdominal wound closure	CP (cutting point)

TAPER NEEDLES IN UROLOGY

Needle	Arc	Size	Tissue
RB/CV (renal bypass / cardiovascular)	½ circle	17 mm — smallest	Most delicate: vessel walls, mucosal surfaces
SH (small half)	½ circle	26 mm — thicker, larger diameter than RB	Intermediate
CT (circle taper)	½ circle	36 mm — greatest diameter	Toughest tissue

Needle	Arc	Size	Tissue
UR/GU (urology / genitourinary)	5/8 circle	—	Deep, narrow spaces , e.g. urethral anastomotic sutures in the deep pelvis

- Two scales: **size rises with tissue toughness** (RB → SH → CT); **arc rises with depth** — UR/GU is the 5/8-circle needle for the deep pelvis.

Concept — the needle point is a trade-off between ease of passage and tissue trauma: **taper stretches** (saves tissue), **cutting pierces** (for dense tissue), **blunt protects the operator**; the reverse-cutting needle moves the edge outward so the wound does not tear toward the edge.

Needle Types, Sizes, and Uses

Free Versus Swaged Needles

- **Swaged**: suture pre-attached — **far more common**, faster, and **thinner** (no eyelet) so less trauma. **Free**: suture threaded through an eyelet — **thicker** to accommodate the eyelet, so more trauma.
- **Use of a free needle**: after a swaged needle has been passed one way, a free needle is attached to the needleless end to pass it back the opposite way. Alternative: **double-armed swaged sutures**, available only in limited needle and material combinations (mainly vascular Prolene).
- Swaged needles are either **permanently attached** (the suture must be cut to free the needle) or **"pop-off"** (released by traction), which saves time between throws before a knot is tied.

Needle Thickness and Diameter

- Needle thickness and diameter **roughly match suture calibre**: heavier suture needing more tensile strength rides on a thicker, larger-diameter needle, so that **the needle hole approximates the suture diameter** and the suture passes easily.
- **Smaller circular diameter** → **precision** in fine suturing; **larger diameter** → **force** to penetrate dense tissue where precision matters less.
- Small variations are described as **fine, medium or heavy**: **medium** for most applications; **fine** for delicate tissue such as bowel or urothelium; **heavy** ("hernia needles") for very dense or scarred tissue where a medium needle may bend or break.

SUTURE MATERIAL

- Three axes of classification: **absorbable vs non-absorbable**, **natural vs synthetic**, and **braided vs monofilament**. Most sutures come in a range of sizes.
- **Natural sutures** (chromic gut, silk) provoke a **greater inflammatory reaction** than synthetic ones.
- **Braided vs monofilament** — the governing pair:
 - **Braided**: easier to handle, **resists memory**, **less knot unravelling**; but **harder to pass** through tissue and more inflammatory.
 - **Monofilament**: **passes easily**, **less inflammation**; more memory, knots less secure.

TABLE 14.1 PROPERTIES OF COMMONLY USED SUTURE MATERIALS

Suture	Origin	Construction	Sizes	Tensile strength	Absorption complete by
ABSORBABLE					
Chromic gut	Natural	Monofilament	7/0–3	21–28 days	90 days
Plain gut	Natural	Monofilament	6/0–0	7–10 days	70 days
Monocryl	Synthetic	Monofilament	6/0–1		91–119 days

Suture	Origin	Construction	Sizes	Tensile strength	Absorption complete by
				1 wk 50–70%; 2 wk 20–40%	
Caprosyn	Synthetic	Monofilament	6/0–1	5 days 60%; 10 days 20–30%	56 days
Biosyn	Synthetic	Monofilament	6/0–1	2 wk 75%; 3 wk 40%	110 days
Vicryl	Synthetic	Braided (8/0–3); monofilament (10/0–9/0)	10/0–3	2 wk 75%; 3 wk 40–50%	56–70 days
PDS	Synthetic	Monofilament	7/0–2	2 wk 60–80%; 6 wk 35–60%	182–238 days
Maxon	Synthetic	Monofilament	7/0–1	2 wk 75%; 6 wk 25%	180 days
NON-ABSORBABLE					
Prolene	Synthetic	Monofilament	10/0–2	Indefinite	—
Mersilene	Synthetic	Braided (6/0–1); monofilament (11/0–10/0)	11/0–1	Indefinite	—
Nylon	Synthetic	Monofilament	11/0–2	Gradual loss over time	—
Silk	Natural	Braided	7/0–2	Gradual loss over time	—
Stainless steel	Synthetic	Monofilament	5/0–7	Indefinite	—

From Hartman C, Kavoussi LR: Handbook of surgical technique, 2018, Elsevier, Table 7.1; Ethicon 2011.

- What the table is for: matching **how long the wound needs support** to how long the suture provides it. Strength falls into four tiers: **about 1–2 weeks** (plain gut, Caprosyn, Monocryl) → **2–4 weeks** (Vicryl, Biosyn, chromic gut) → **6 weeks and beyond** (PDS, Maxon) → **indefinite** (Prolene, Mersilene, steel).
- The trap: **loss of strength precedes absorption by weeks to months** (Vicryl strong at 2–3 weeks, gone by 70 days; PDS gone by 6–8 months); and **nylon and silk are non-absorbable yet lose strength gradually**.

Absorbable Sutures

Chromic Gut

- **Serosal lining of sheep or bovine intestine treated with chromium salt**, which slows absorption. Monofilament in form, yet **drags through tissue** — may be coated with petroleum jelly.
- For sites where **rapid dissolution matters and strength does not: circumcision, scrotal skin**. Provokes one of the greatest inflammatory reactions of all sutures.

Plain Gut

- Same origin, **not chromium-treated** → even shorter strength; similar uses, but **used less**.

Monocryl (Poliglecaprone 25), Caprosyn and Biosyn

- All three are **synthetic monofilaments for subcuticular skin closure**: pass easily, minimal inflammation. They differ in one feature each:

Suture	Skin-closure strength	Absorption	Distinguishing feature
Monocryl	7–14 days	Reference	The reference skin monofilament
Caprosyn	Shorter — up to 10 days	About half Monocryl's time	Faster absorption → theoretically less inflammation and infection
Biosyn	Greater than either	Similar to Monocryl	Strongest of the three

Vicryl (Polyglactin 910)

- A **braided synthetic**, among the **most used sutures in urologic surgery**: reapproximation of **fat, muscle and fascia**, **ligation of small vessels**, and **anastomoses**.
- Adequate support (**2–3 weeks**) with **early absorption (≤ 70 days)** → less infection and inflammation than longer-lasting materials.

Polydioxanone (PDS) and Maxon (Polyglyconate)

- Both **synthetic monofilaments for wound and fascial closure**, both available as a **double-stranded loop** for extra strength.
- **PDS**: passes easily; **security beyond 6 weeks**, yet absorbed — less inflammation and infection than permanent material; the choice for **prolonged fascial strength**. **Maxon**: security **up to 6 weeks**, absorbed within 6 months.

Nonabsorbable Sutures

Suture	Construction	Strength	Use	Cautions
Prolene (polypropylene)	Monofilament	Indefinite; does not degrade	High strength over a prolonged period	Never in contact with the urinary tract — stone nidus ; smooth and highly plastic, so knots unravel — extra throws needed
Mersilene (polyester)	Braided	Indefinite	High-stress sites needing prolonged strength; holds knots better than Prolene	Not in the urinary tract (stones); not in infection — braid favours bacterial adherence
Nylon (polyamide)	Monofilament	Lost gradually	Drain suture : plastic, minimal inflammation — easy, painless removal	—
Silk	Braided (natural)	Lost gradually, but sufficient	Handles well, knots secure; marks tissue where re-entry is anticipated — e.g. orchidectomy before possible retroperitoneal lymph node dissection (RPLND) and gonadal vessel resection	Natural — more inflammatory

Barbed Suture

Suture	How barbs are made	Strength	Anchor	Uses and limits
V-Loc	Cut into the monofilament	Variants of 14 days, 21 days, permanent	Loop at the end; needle passed through it	Knotless — valuable in laparoscopic and robotic suturing; cutting weakens the filament, so not for fascial closure
Quill; Stratafix Spiral	Barbed monofilament	Fast-absorbing options	—	Soft tissue and skin; double-armed, bidirectional barbs (holding toward the centre) close long incisions rapidly subcuticularly

Suture	How barbs are made	Strength	Anchor	Uses and limits
Stratafix Symmetric PDS	Moulded onto the monofilament	Significant; PDS — absorbable yet prolonged	Flat tab — no knot, no loop	Larger sizes permitted for fascial closure, including midline incisions

Concept — a barbed suture's strength depends on how the barbs are made: **cut barbs weaken the filament and exclude fascia; moulded barbs do not**, which is why Stratafix Symmetric is the only barbed suture strong enough to close fascia.

SUTURING TECHNIQUES

- Technique balances **mechanical stress, operating time, infection risk and cosmesis**, and is matched to tissue.

Skin Closure

- Dermis and epidermis are reapproximated to **restore the barrier** unless drainage is required (e.g. pre-existing infection); some wounds carry tension, others need only edge approximation to limit scarring.
- A correctly approximated, uninfected **epidermis seals within 24–48 hours** but lacks dermal strength — so the suture must hold, or remain before removal or absorption, **until the dermis is strong** (which varies with body area).
- **Running vs interrupted** (applies to every technique): running saves knot-tying time, but a **break anywhere can open the entire incision**.

Technique	Needle path	Advantages	Disadvantages and notes
Simple (Fig. 14.1E)	Full thickness of epidermis and dermis into hypodermis, each side	Easy, fast	Part of suture exposed at the non-sterile surface; monofilament preferred — braid lets bacteria travel along it; external or buried knots; interrupted or running
Figure-of-eight (F)	As simple, then a second pass through each side further along before tying	More redundancy with half the knots of interrupted; central compression gives haemostasis of small vessels	—
Vertical mattress (G)	First pass far from the edges , return pass close to the edges ; knot on one side	Skin eversion; less tension at the edges	Both pass full thickness twice on each side
Horizontal mattress (H)	Travel along the incision before the return pass; knot on one side, forming a loop		
Subcuticular (I)	Within superficial dermis just below the cuticle	No external knots or sutures , less infection, better cosmesis	Absorbable suture required ; knots buried; slightly longer
Deep dermal	Just below the epidermis, knot buried	Excellent approximation, especially under tension ; takes tension off the epidermis → less deformation, better cosmesis	Often combined with running subcuticular, Dermabond, or both

- **Subcuticular buried knot** — **sequence**: first pass **starts deep** → exits in the subcuticular layer → next pass from superficial layer to deep hypodermis → both ends exit the **same side** → knot falls into the

hypodermis; same at the far end. **Barbed Monocryl** may be used; with two surgeons a **double-armed suture started in the middle** is run in opposite directions; if the skin is not under tension, a **bidirectional barbed suture needs no knots**.

- **Deep dermal buried knot — sequence:** enter in the hypodermis → exit in the dermis below the cuticle → opposite side, enter just below the cuticle → exit in the hypodermis → needle exits on the **same side as the free end** → knot drops into the hypodermis out of view.

Concept — *every skin technique trades speed against security and surface exposure against cosmesis: burying the suture (subcuticular, deep dermal) removes surface exposure and epidermal tension at the cost of time.*

Fascial Closure

- **Fascial defects of 1 cm or larger are closed** after abdominal surgery.
- Fascia is **poorly vascularised and heals slowly**, so it needs **longer tensile strength**; some applications benefit from **permanent sutures**. Surgeon technique determines both **avoidance of visceral injury** and **incisional hernia risk**.
- **STITCH trial** (landmark): midline laparotomy randomised to **small bites (5 mm from the edge, every 5 mm)** vs **large bites (1 cm from the edge, every 1 cm)**. Large bites were **faster (10 vs 14 minutes)** but gave **more incisional hernia at 1 year (21% vs 13%; P = 0.022)**. Small bites are not yet universal standard, but **spacing greater than 1 cm should be avoided**.

	Running (continuous)	Interrupted
Material	Large-diameter PDS or permanent (Prolene); Stratafix is the only barbed suture strong enough	Slow-absorbing or permanent, and Vicryl also acceptable ; size matched to anticipated stress
Patterns	Single continuous strand	Simple interrupted, mattress, figure-of-eight
Advantage	Fast , few knots	Security if any single suture fails or tears through
Disadvantage	Theoretical loosening of the whole closure if any point fails → long-term strength essential	More knot-tying
Evidence	No increase in incisional hernia ; a systematic review found fewer hernias with running closure provided slow-absorbing suture was used —	

- **Laparoscopic port sites greater than 8 mm** (e.g. assistant port) — suture passer, under direct vision: pass suture through fascia into the abdomen on one side → passer through fascia on the opposite side → grasp suture, pull it out beneath the skin into the incision → repeat for multiple strands → tie knots external to the fascia.

Concept — *fascia heals slowly, so duration of strength governs every choice: a running closure is safe only when its material outlasts healing, and small, closely spaced bites distribute tension better than large ones.*